

① Find a Regular Expression corresponding to each of the following subsets of $\{a, b\}^*$. (3 marks)

① set of all strings with length 5 exactly and next to last symbol is 'b'.

→ alphabet $\Sigma = \{a, b\}$

length is 5, next to last symbol is 'b'

so,

first 3 positions and last (5th) position can be a or b i.e. $a+b$

$$\therefore \text{R.E.} = (a+b)(a+b)(a+b)b(a+b)$$

$$= (a+b)^3 b (a+b)$$

② set of all strings ends with sub-string 'abb'

→ alphabet $\Sigma = \{a, b\}$

 abb

here, any sequence of a and b can come before abb.

$$\therefore \text{R.E.} = (a+b)^* abb$$

③ set of all strings having 'a' as first symbol and 'b' as last symbol.

→ alphabet $\Sigma = \{a, b\}$

$$\therefore \text{R.E.} = a(a+b)^* b$$

②

Write Regular Expressions corresponding to each of the following subsets of $\{0,1\}^*$ (4 Marks)

(i) The language of all strings containing both 101 and 010 as substrings.

(ii) The language of all strings that do not end with 01.

① $L = \{ \text{strings containing both 101 and 010 as substrings} \}$

→ these two substrings 101 and 010 can appear in any order.

so, there are only two possibilities:

(i) 101 comes first, then later 010

(ii) 010 comes first, then later 101

case 1: 101 appears before 010

∴ string structure can be:

anything + 101 + anything + 010 + anything

now, "anything" comes from $\Sigma = \{0,1\}$
any sequence of 0s and 1s

∴ anything = $(0+1)^*$

so:

$(0+1)^* 101 (0+1)^* 010 (0+1)^*$

case 2: 010 appears before 101

∴ $(0+1)^* 010 (0+1)^* 101 (0+1)^*$

combine both cases: Final RE

$(0+1)^* 101 (0+1)^* 010 (0+1)^* + (0+1)^* 010 (0+1)^* 101 (0+1)^*$

② $L = \{ \text{strings that do not end with } 01 \}$

→ possible valid endings :

00, 10, 11, single 0, single 1, ϵ

so, valid endings are :

ends with 0 or

ends with 11 or

ends with ϵ or single 1

convert in RE :

strings ending with 0 : $(0+1)^* 0$

strings ending with 11 : $(0+1)^* 11$

short valid strings : ϵ or 1

combine all : Final RE

$$\underline{\epsilon + 1 + (0+1)^* 0 + (0+1)^* 11}$$

③ Find a regular expression of following subsets of $\{0, 1\}^*$ (4 Marks)

(i) The language of all strings that begin or end with 00 or 11.

(ii) The language of all strings ending with 1 and not containing 00.

① $L = \{ \text{strings begin or end with } 00 \text{ or } 11 \}$

4 case : begins with 00 → $00(0+1)^*$

begins with 11 → $11(0+1)^*$

ends with 00 → $(0+1)^* 00$

ends with 11 → $(0+1)^* 11$

combine all cases using union (+)

Final Regular Expression :

$$\underline{00(0+1)^* + 11(0+1)^* + (0+1)^* 00 + (0+1)^* 11}$$

② $L = \{\text{strings ending with 1 and do not contain 00}\}$

as '00' is restricted,

if we take 0, what can come next? only 1.

so, after every 0, there must be 1.

$\therefore$ valid strings: 1, 01, 101, 111, 0101, 101101, 01101

any string made from 1 and 01

$$\therefore \text{R.E.} = (1+01)^* \underbrace{(1+01)}_{\text{why?}}$$

because $\rightarrow$ kleene star (*) means:

$$(1+01)^* \rightarrow \epsilon + (1+01) + (1+01)(1+01) + \dots$$

$\rightarrow$ given language says: strings ending with 1

so, according to kleene star, does empty string

ϵ end with 1? $\rightarrow$ No.

so, we need to add $(1+01)$ along with $(1+01)^*$

$$\therefore \underline{\text{final R.E.} = (1+01)^* (1+01)}$$

④

Write regular expressions for the following. (4 Marks)

(i) Binary numbers that are multiple of 2.

(ii) Strings of a's and b's with no consecutive a's.

(iii) Strings of a's and b's containing consecutive a's.

① Binary numbers that are multiple of 2.

Binary numbers are formed using 0 & 1.

$$\therefore \Sigma = \{0, 1\}$$

multiple of 2 = 2, 4, 6, 8, 10, 12, ...

↓ ↓ ↓ ↓ ↓
10 100 110 1000 1010

→ Binary numbers which are multiple of 2 ends with 0

$$\therefore \text{R.E.} = (0+1)^* 0$$

② strings of a's and b's with no consecutive a's

→ avoid 'aa' in string.

$L = \{ \epsilon, a, b, ab, ba, bab, aba, babab, bbbab, bbbabbb, abab, baba, \dots \}$

$a^* \rightarrow \epsilon, a, aa, \times$ $b^* \rightarrow \epsilon, b, bb, bbb \checkmark$

$(ab)^* \rightarrow \epsilon, ab, abab, ababab, \dots \checkmark$

$(ba)^* \rightarrow \epsilon, ba, baba, bababa, \dots \checkmark$

$(ab+ba)^* \rightarrow ab, \cancel{abaa}, \times$

$(ba+ab)^* \rightarrow \epsilon, b, ba, baba, bab, babab, bba, bbbab \checkmark$

$$\therefore \text{R.E.} = (ba+ab)^*$$

but this R.E. won't give string 'a' and ϵ

$$\therefore \text{Final R.E.} = (\epsilon+a)(ba+ab)^*$$

③ strings of a's and b's with consecutive a's.

$L = \{ \epsilon, aa, abaa, bbaa, aab, baabaaab, \dots \}$

$a^* \rightarrow \epsilon, aa \checkmark$

$(a+b)^* \rightarrow \epsilon, aa, aab, b, bb \checkmark$

$$\text{R.E.} = (a+b)^* aa (a+b)^*$$

5

Construct the regular expressions for the following languages. (4 Marks)

$L_1 = \{\text{Where the no. of 'a' is odd}\}, \Sigma = \{a, b\}$

$L_2 = \{\text{Where every string starts with '0' and of even length}\}, \Sigma = \{0, 1\}$

① $L_1 = \{\text{no of a's is odd}\}$

$L_1 = \{a, aaa, baaa, aaab, bbba, bba, bab, babaa, \dots\}$

odd $\rightarrow 1, 3, 5, 7, \dots$

$\begin{matrix} \downarrow & \downarrow & \downarrow \\ a & aaa & \rightarrow aaaaa \end{matrix}$

$\rightarrow$ if one 'a', then we can put 'b' anywhere.

$\therefore ab, ba, abb, bbabb, \dots$

$\therefore R.E. = b^*ab^*$

for odd no of a's we need to add 2'a in one 'a' $\rightarrow 1+2=3$

$\therefore R.E. = (b^*ab^*)(b^*ab^*ab^*)$

what about other odd no of a's $\rightarrow$ i.e. 5, 7,

$\therefore 1'a' + 2'a' = 3'a'$

$1'a' + 4'a' = 5'a'$

$1'a' + 6'a' = 7'a'$

$\therefore (b^*ab^*ab^*)^*$

final R.E. = $(b^*ab^*)(b^*ab^*ab^*)^*$

② $L_2 = \{\text{strings start with '0' and even length}\}$

$L_2 = \{01, 00, 0100, 0000, 011101, \dots\}$

even length : 2, 4, 6, 8,

$0_ , 0_ _ _ , 0_ _ _ _ , 0_ _ _ _ _ _$

$0(0+1) \rightarrow 0(0+1)((0+1)(0+1))^*$

Final R.E. = $0(0+1)((0+1)(0+1))^*$

6

Construct the regular expressions for the following languages. (4M, w22)

$L_1 = \{\text{Where every string starts with 'b' and does not contain 2 consecutive a's}\}$, $\Sigma = \{a, b\}$

$L_2 = \{\text{Where every string starts with '1' and of odd length}\}$, $\Sigma = \{0, 1\}$

① $L_1 = \{\text{string starts with 'b', don't contain 2 consecutive a's}\}$

$$L_1 = \{b, bbab, babb, bbbab, \dots\}$$

$$b^* = \epsilon, b, bb, \dots \checkmark$$

along with b , there can be any no of 'b's,
only one a , or any number of ab or ba

$$\therefore \text{R.E.} = b(b+ab)^*(\epsilon+a)$$

strings = $b, ba, bbbb, bab, babab, bba$

→ we can't write $(b+ba)$. because it will
lead to 2 consecutive a's.

$$\underline{\text{ex}} \quad b(b+ba)^*(\epsilon+a)$$

$$= \underset{\checkmark}{b}, \underset{\checkmark}{bb}, \underset{\checkmark}{bba}, \underset{\times}{bbba}$$

② $L_2 = \{\text{strings starts with '1' and odd length}\}$

$$L_2 = \{1, 101, 100, 10100, 11001, \dots\}$$

$$\text{R.E.} = 1((0+1)(0+1))^*$$

